

Important Questions for Class 12

Mathematics

Chapter 12 – Linear Programming

Long Answer Type Questions

6 Marks

1. Solve the following L.P.P. graphically

Minimise and maximise $z = 3x + 9y$

Subject to the constraints $x + 3y \leq 60$

$$x + y \geq 10$$

$$x \leq y$$

$$x \geq 0, y \geq 0$$

Ans:

Maximize $Z = 3x + 9y$

Minimize $Z = 3x + 9y$

Subject to,

$$x + 3y \leq 60$$

$$x + y \geq 10$$

$$x \leq y$$

$$x \geq 0, y \geq 0$$

$$x + 3y \leq 60$$

X	0	60
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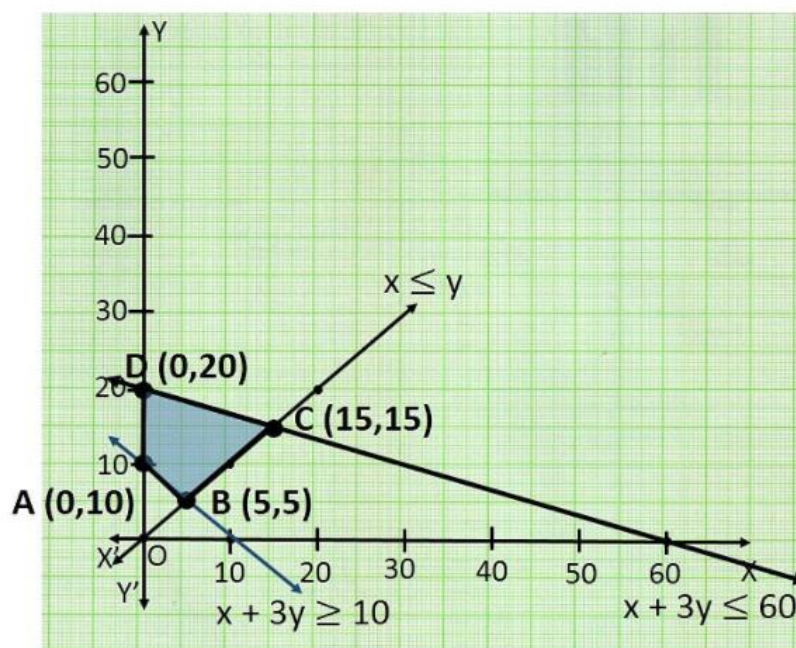
y	20	0
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$$x + y \geq 10$$

X	0	10
y	10	0

$$x \leq y$$

X	0	20
y	0	20



Corner points	Value of Z
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(0,10)	90
(5,5)	60
(15,15)	180
(0,20)	180

$\therefore Z = 60$ is minimum at (5,5)

Also,

$Z = 60$ is maximum at two points (15,15) & (0,20)

$\therefore Z = 180$ is maximum at all points joining (15,15) & (0,20).

2. Determine graphically the minimum value of the objective function $z = -50x + 20y$, subject to the constraints

$$2x - y \geq -5$$

$$3x + y \geq 3$$

$$2x - 3y \leq 12$$

$$x \geq 0, y \geq 0$$

Ans:

Minimum $Z = -50x + 20y$

Subject to,

$$2x - y \geq -5$$

$$3x + y \geq 3$$

$$2x - 3y \leq 12$$

$$x \geq 0, y \geq 0$$

$$2x - y \geq -5$$

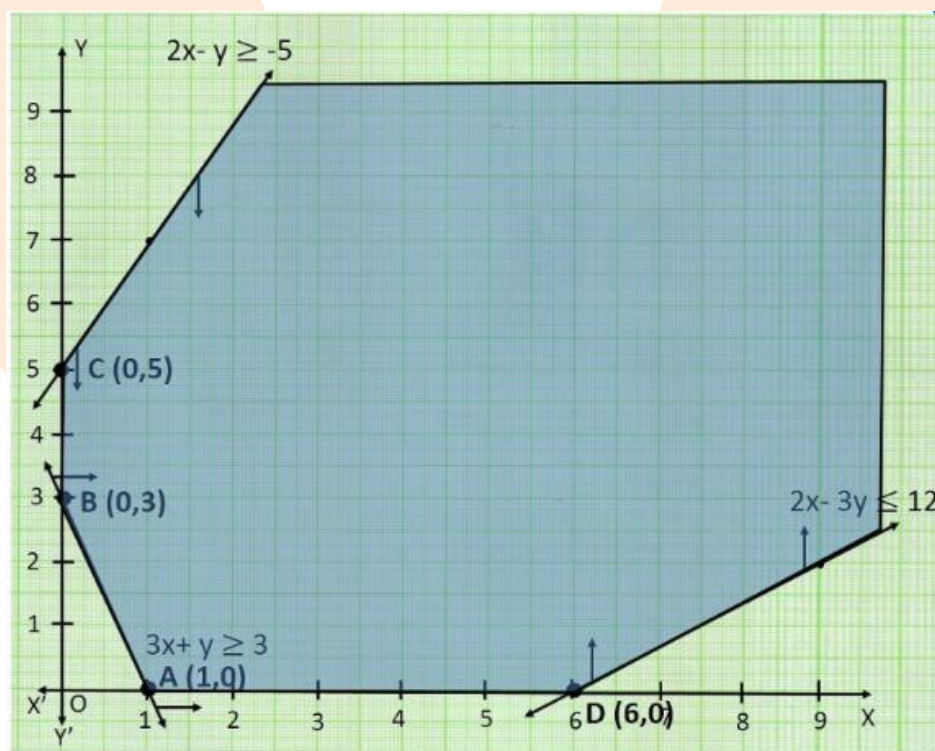
x	0	1
y	5	7

$$3x - 3y \geq 3$$

X	1	0
y	0	3

$$2x - 3y \leq 12$$

X	6	9
y	0	2



Corners points	Value of Z
(0,5)	100
(0,3)	60
(1,0)	-50
(6,0)	-300

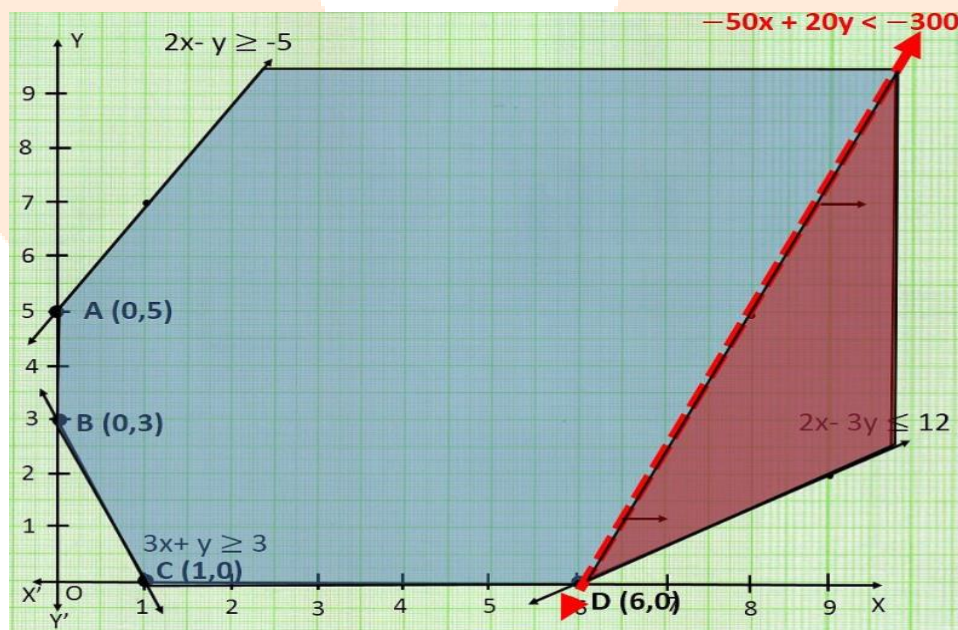
As the region that is feasible is unbounded.

Hence -300 may or may not be the minimum value of Z .

To check this, we graph inequality.

$$-50x + 20y < -300$$

X	6	8
y	0	5



As, there are some common points between the feasible region the inequality.

Hence, 300 is not the minimum value of Z .

$\therefore Z$ has no minimum value.

3. Two tailors A and B earn Rs. 150 and Rs. 200 per day respectively. A can stitch 6 shirts and 4 pants per day, while B can stitch 10 shirts and 4 pants per day. Formulate the above L.P.P. mathematically and hence solve it to minimize the labour cost to produce at least 60 shirts and 32 pants.

Ans:

Let Tailor A work for x days

Tailor B work for y days

Given that

Two tailors A and B earn 150 and 200 per day respectively.

And, we need to minimize labour cost

So, our equation will be

$$\therefore Z = 150x + 200y$$

Now, according to question

A can stitch 6 shirts and 4 pants per day, while B can stitch 10 shirts and 4 pants per day. We need to minimize the labour cost to produce at least 60 shirts and 32 pants and solve it graphically

Shirts

Pants

$$\begin{aligned} \text{Min Shirts} &= 60 \\ 6x + 10y &\geq 60 & 4x + 4y &\geq 32 \\ 3x + 5y &\geq 30 & x + y &\geq 8 \end{aligned}$$

$$\text{Min Pants} = 32$$

Also,

$x \geq 0, y \geq 0$ Combining all constraints:

$$\text{Min } Z = 150x + 200y$$

Subject to constraints,

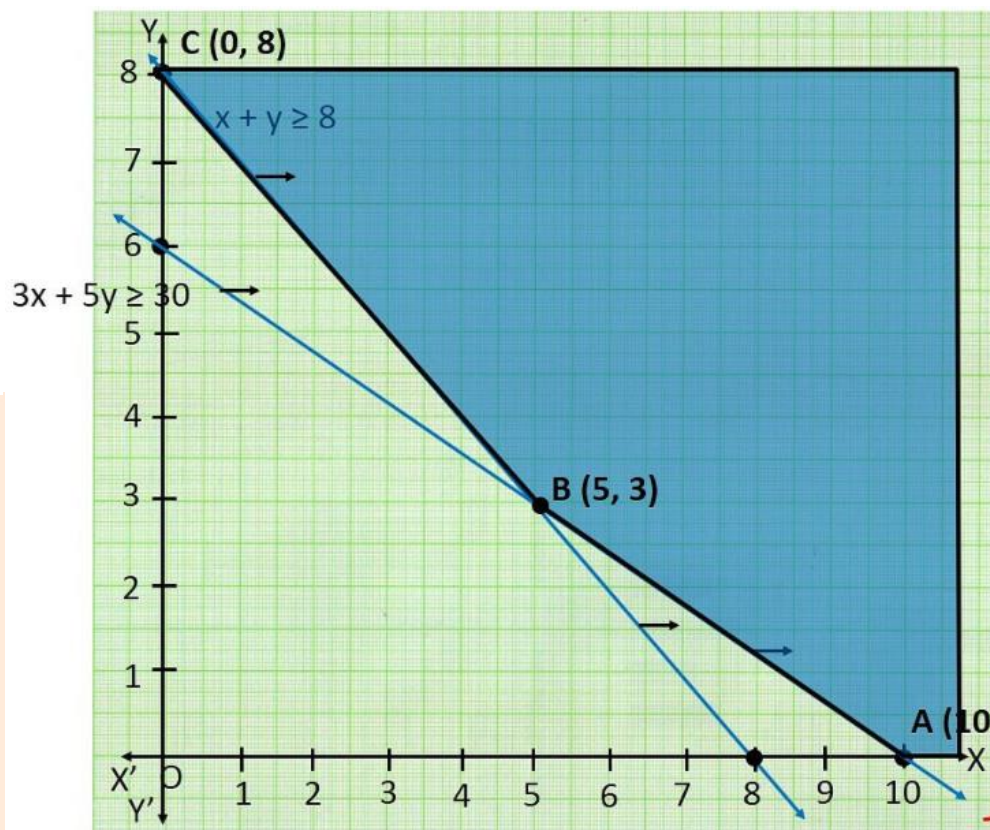
$$\begin{aligned} 3x + 5y &\geq 30 \\ x + y &\geq 8 \\ \& x \geq 0, y \geq 0 \end{aligned}$$

$$3x + 5y \geq 30$$

X	10	0
y	0	6

$$x + y \geq 8$$

X	8	0
y	0	8



Corner Points	$Z = 150x + 200y$
A (10,0)	1500
B (5,3)	$150(5) + 200(3) = 1350$
C (0,8)	1600

As the region that is feasible is unbounded.

Hence 1350 may or may not be the minimum value of Z .

To check this, we graph inequality.

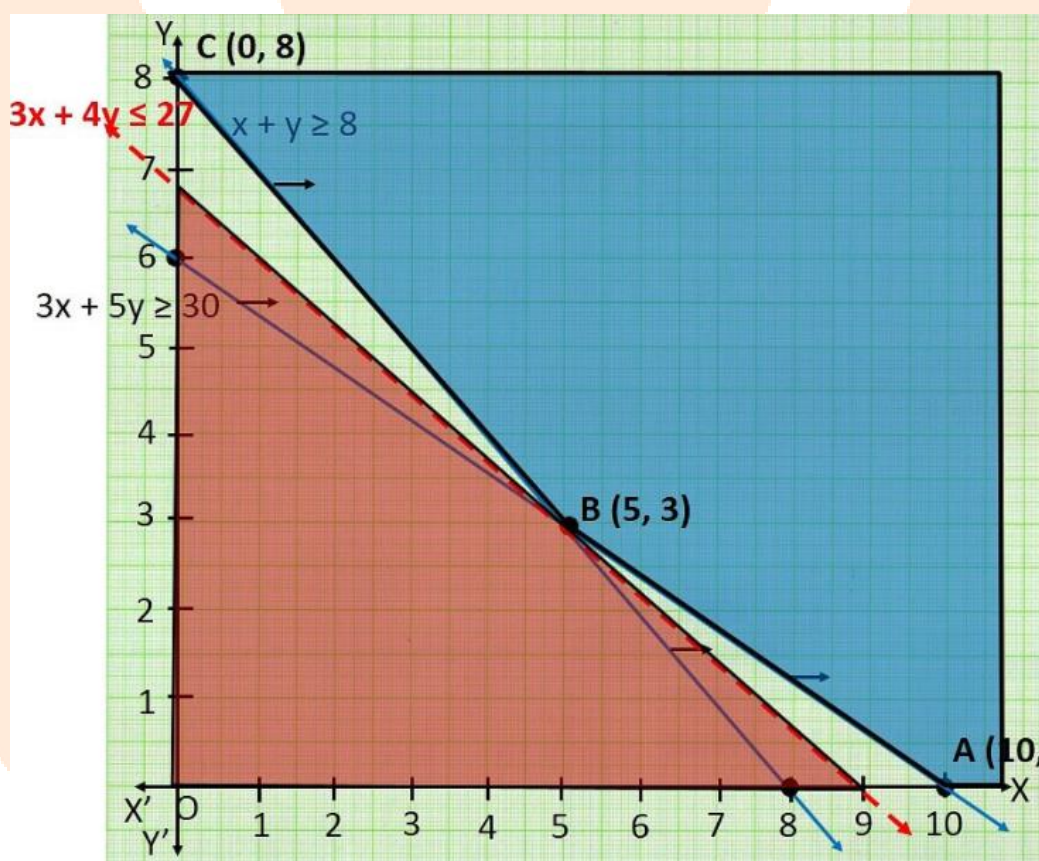
$$150x + 200y \leq 1350$$

$$\text{i.e. } 3x + 4y \leq 27$$

$$3x + 4y \leq 27$$

$$3x + 4y \leq 27$$

X	9	0
y	0	6.75



As there is no common point between the feasible region & the inequality.

Hence, 1350 is the minimum value of Z .

Hence, cost will be minimum if

Number of days tailor A works for = 5

Number of days tailor B works for = 3

Minimum Cost = Rs 1350 at 5 days of A and 3 days of B .

4. There are two types of fertilizer's A and B . A consists of 10% nitrogen and 6% phosphoric acid and B consists of 5% nitrogen and 10% phosphoric acid. After testing the soil conditions, a farmer finds that he needs at least 14 kg of nitrogen and 14 kg of phosphoric acid for his crop. If A costs Rs. 6 kg and B costs Rs. 5 kg, determine how much of each type of fertilizer should be used so that nutrient requirements are met at minimum cost. What is the minimum cost?

Ans: Let the fertilizer F_1 requirement be x kg

fertilizer F_2 requirement be y kg

Fertilizer	Nitrogen	Phosphoric Acid	Quantity	Cost
F1	10 %	6%	X kg	Rs 6
F2	5%	10%	Y kg	Rs 5
Total	14 kg	14kg		

According To questions:

Nitrogen	Phosphoric Acid
Nitrogen in F1 = 10%	Acid in F1 = 6%
Phosphoric in F2 = 5%	Acid in F2 = 10%

Minimum Available = 14 kg	Minimum Available = 14 kg
10% of x + 5% of y $\geq$ 14	6% of x + 10% of y $\geq$ 14
2x + y $\geq$ 280	3x + 5y $\geq$ 700

Also, $x \geq 0, y \geq 0$

As we need to Minimize the cost,

hence the function used here is Minimize z .

Cost of fertilizer F_1 = Rs 6 kg

Cost of fertilizer F_2 = Rs 5 kg

Minimize $z = 6x + 5y$

Combining all Constraints:

Min $Z = 6x + 5y$

Subject to constraints

$$2x + y \geq 280$$

$$3x + 5y \geq 700$$

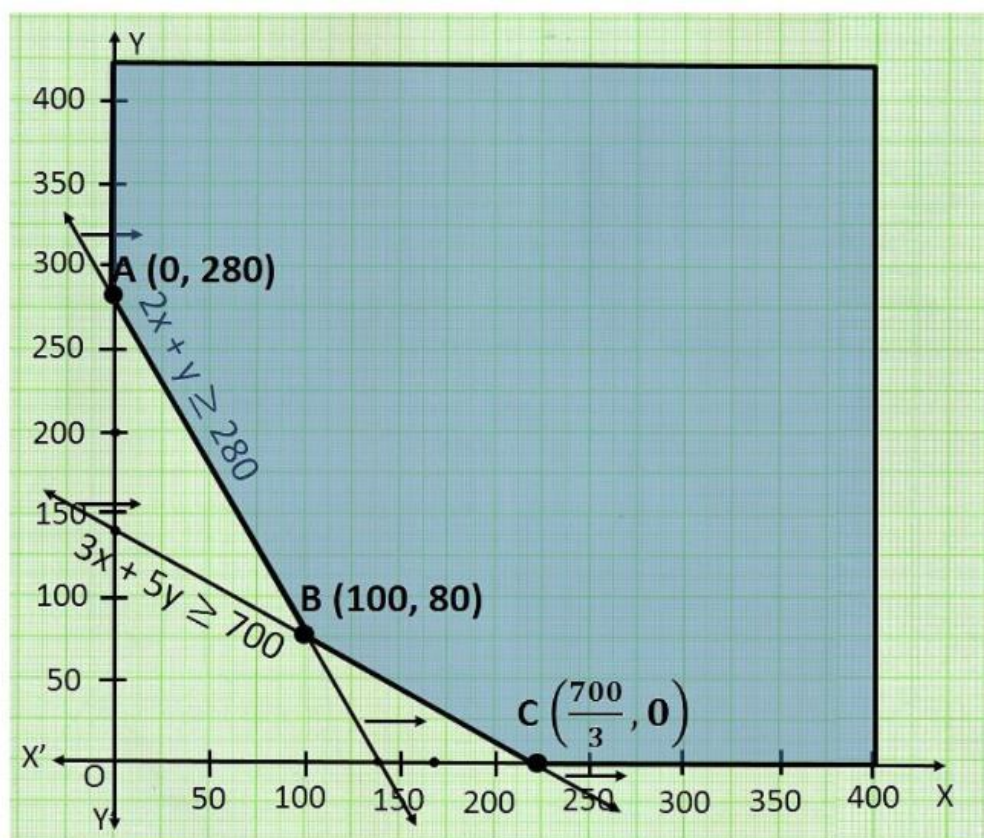
$$x, y \geq 0$$

$$2x + y \geq 280$$

X	140	0
y	0	280

$$3x + 5y \geq 700$$

X	0	$\frac{700}{3}$
Y	140	0



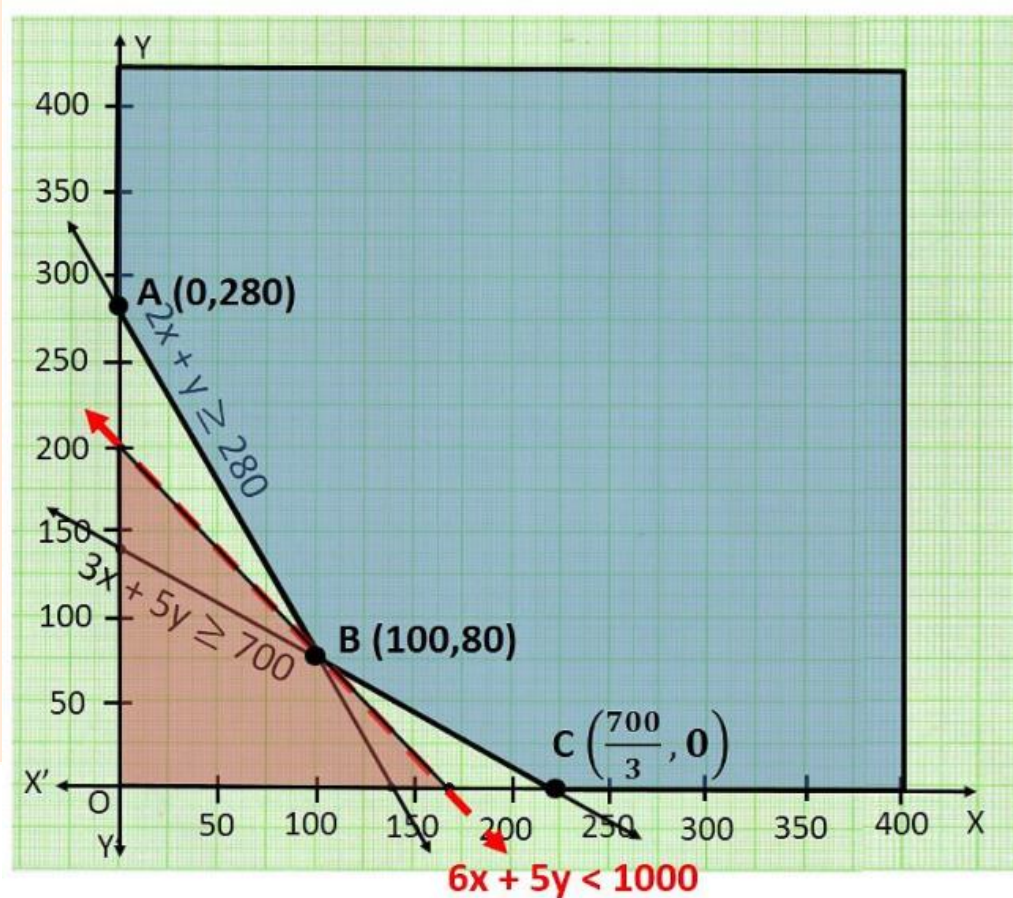
Corner Points	Value of Z
(0,280)	1400
(100,80)	1000
($\frac{700}{3}$,0)	1400

As the feasible area is unbounded

Hence 1000 may or may not be the minimum value of Z .

For this, we need to graph inequality

X	500/3	0
Y	0	200



Since, there is no common point between the feasible region and

$$6x + 5y < 1000$$

Hence the cost will be minimum, if

Fertilizer F_1 used = 100kg

Fertilizer F_2 used = 80kg

Minimum Cost = Rs 1000.

5. A man has Rs. 1500 to purchase two types of shares of two different companies S1 and S2. Market price of one share of S1 is Rs 180 and S2 is Rs. 120. He wishes to purchase a maximum of ten shares only. If one share of type S1 gives a yield of Rs. 11 and of type S2 yields Rs. 8 then how much shares of each type must be purchased to get maximum profit? And what will be the maximum profit?

Ans:

Let the S1 purchased be x and S2 purchased be y

Then, $180x + 120y \leq 1500$

Also, $x + y \leq 10$

$x \geq 0$ and $y \geq 0$

$Profit = 11x + 8y$

Now, if you will draw the graph using the 4 constraint equations mentioned above, the common region would be a closed surface with A (0,0), B (8.33,0), C (5,5), D (0,10)

Hence, the profit would be maximum at one of the vertices only

$Profit \text{ at } A (0,0) = 11(0) + 8(0) = 0$

$Profit \text{ at } B (8.33,0) = 11(8.33) + 8(0) = 91.63$

$Profit \text{ at } C (5,5) = 11(5) + 8(5) = 95$

$Profit \text{ at } D (0,10) = 11(0) + 8(10) = 80$

Hence, the maximum profit is Rs 95 at point C and hence, the man should purchase 5 Share of S1 and S2 to maximise his profit.

6. A company manufactures two types of lamps say A and B. Both lamps go through a cutter and then a finisher. Lamp A requires 2 hours of the cutter's time and 1 hours of the finisher's time. Lamp B requires 1 hour of cutter's and 2 hours of finisher's time. The cutter has 100 hours and finishers has 80 hours of time available each month. Profit on one lamp A is Rs. 7.00 and on one lamp B is Rs. 13.00. Assuming that he can sell all that he produces, how many of each type of lamps should be manufactured to obtain maximum profit?

Ans:

Let the company manufacture x cutters of type A and y finishers of type B.

Therefore,

$$x \geq 0 \text{ and } y \geq 0$$

The given information can be compiled in a table as follows.

	Lamp A	Lamp B	Availability
Cutter	2	1	100
Finisher	1	2	80

The profit on type A cutters is Rs 5 and on type B finishers is Rs 6. Therefore, the constraints are

$$2x + y \leq 100$$

$$x + 2y \leq 80$$

Total profit, $z = 7x + 13y$

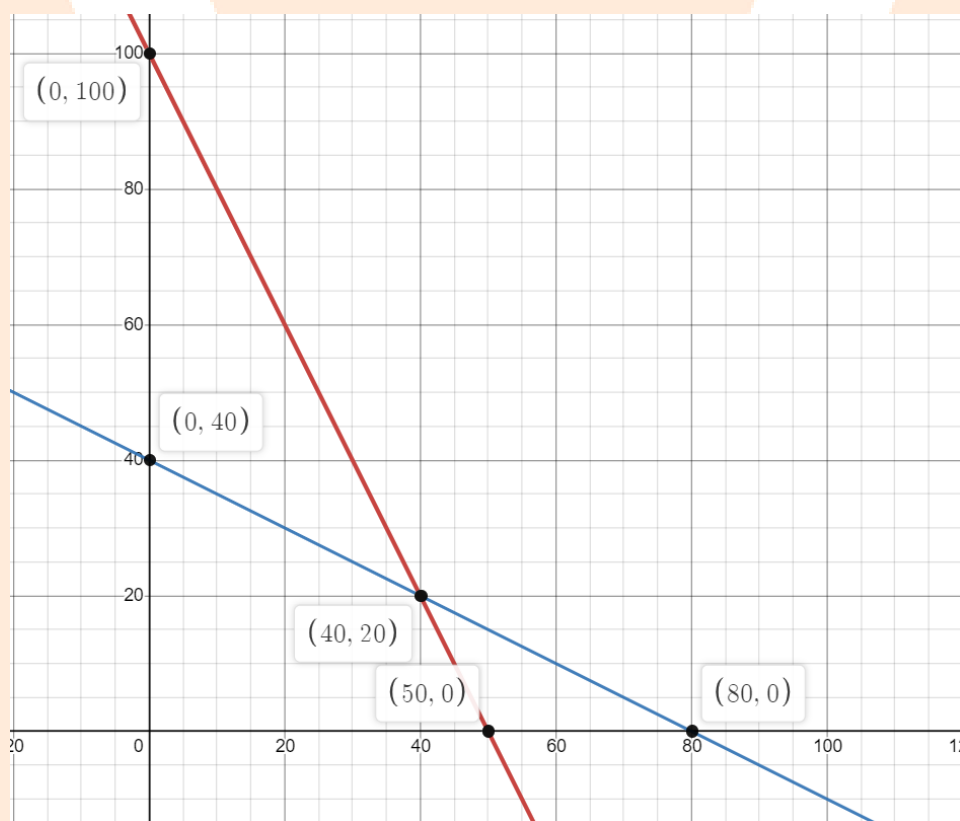
The mathematical formulation of the given problem is Maximize $z = 7x + 13y \dots (1)$

subject to the constraints, $2x + y \leq 100 \dots (2)$

$$x + 2y \leq 80 \dots$$

$$x, y \geq 0 \dots (4)$$

The feasible region determined by the system of constraints is as follows.



As per the constraints the feasible solution is the shaded region. Possible points for maximizing Z are $A(0, 40), B(40, 20), C(50, 0)$

$$Z|_A = 7 \times 0 + 13 \times 40 = 520$$

$$Z|_B = 7 \times 40 + 13 \times 20 = 540$$

$$Z|_C = 7 \times 50 + 13 \times 0 = 350$$

Hence profit is maximum of Rs 540 when the dealer purchases 40 Lamp A and 20 Lamp B.

7. A dealer wishes to purchase a number of fans and sewing machines. He has only Rs. 5760 to invest and has space for almost 20 items. A fan and sewing machine cost Rs. 360 and Rs. 240 respectively. He can sell a fan at a profit of Rs. 22 and sewing machine at a profit of Rs. 18. Assuming that he can sell whatever he buys, how should he invest his money to maximize his profit?

Ans:

Let the no. of fans purchased by the dealer = x

and number of sewing machines purchased = y

then the L.P.P. is formulated as

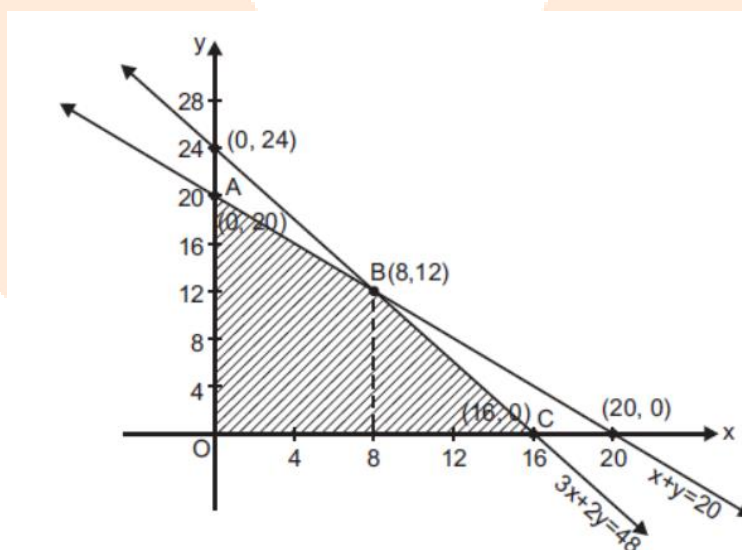
$Z = 22x + 18y$ to be maximised subject to constraints

$$\begin{aligned} x + y &\leq 20 \\ 360x + 240y &\leq 5760 \end{aligned} \quad \dots(i) \text{ [space only for 20 items]}$$

$$\Rightarrow 3x + 2y \leq 48 \quad \dots(ii)$$

$$x \geq 0, y \geq 0 \quad \dots(iii)$$

We plot the graph of the constraints.



As per the constraints the feasible solution is the shaded region. Possible points for maximising Z are $A(0, 20), B(8, 12), C(16, 0)$

$$Z]_A = 22 \times 0 + 18 \times 20 = 360$$

$$Z]_B = 22 \times 8 + 18 \times 12 = 392$$

$$Z]_C = 22 \times 16 + 18 \times 0 = 352$$

Hence profit is maximum of Rs 392 when the dealer purchases 8 fans and 12 sewing machines.

8. If a young man rides his motorcycle at 25 km/h, he has to spend Rs. 2 per km on petrol. If he rides at a faster speed of 40 km/h, the petrol cost increases to Rs. 5 per km. He has Rs. 100 to spend on petrol and wishes to find the maximum distance he can travel within one hour. Express this as L.P.P. and then solve it graphically.

Ans:

Let young man drives x km at a speed of 25 km/ hr. and y km at a speed of 40 km/hr.

Clearly,

$$x, y \geq 0$$

It is given that, he spends Rs 2 per km if he drives at a speed of 25 km/ hr and Rs 5 per km if he drives at a speed of 40 km/ hr. Therefore, money spent by him when he travelled x km and y km are Rs $2x$ and Rs $5y$ respectively.

It is given that he has a maximum of Rs 100 to spend.

Thus, $2x + 5y \leq 100$

Time spent by him when travelling with a speed of 25 km / hr = $\frac{x}{25}$ hr

Time spent by him when travelling with a speed of 40 km / hr = $\frac{y}{40}$ hr

Also, the available time is 1 hour.

$$\frac{x}{25} + \frac{y}{40} \leq 1$$

Or, $40x + 25y \leq 1000$

The distance covered is $Z = x + y$ which is to be maximized.

Thus, the mathematical formulation of the given linear programming problem is

Max $Z = x + y$ subject to

$$2x + 5y \leq 100$$

$$40x + 25y \leq 1000$$

$$x, y \geq 0$$

First, we will convert inequations as follows:

$$2x + 5y = 100$$

$$40x + 25y = 1000$$

$$x = 0 \text{ and } y = 0$$

The region represented by $2x + 5y \leq 100$

The line $2x + 5y = 100$ meets the coordinate axes at $A(50, 0)$ and $B(0, 20)$ respectively. By joining these points, we obtain the line $2x + 5y = 100$. Clearly $(0, 0)$ satisfies the $2x + 5y = 100$. So, the region which contains the origin represents the solution set of the inequation $2x + 5y \leq 100$

The region represented by $40x + 25y \leq 1000$

The line $40x + 25y = 1000$ meets the coordinate axes at $C(25, 0)$ and $D(0, 40)$ respectively. By joining these points, we obtain the line $2x + y = 12$. Clearly $(0, 0)$

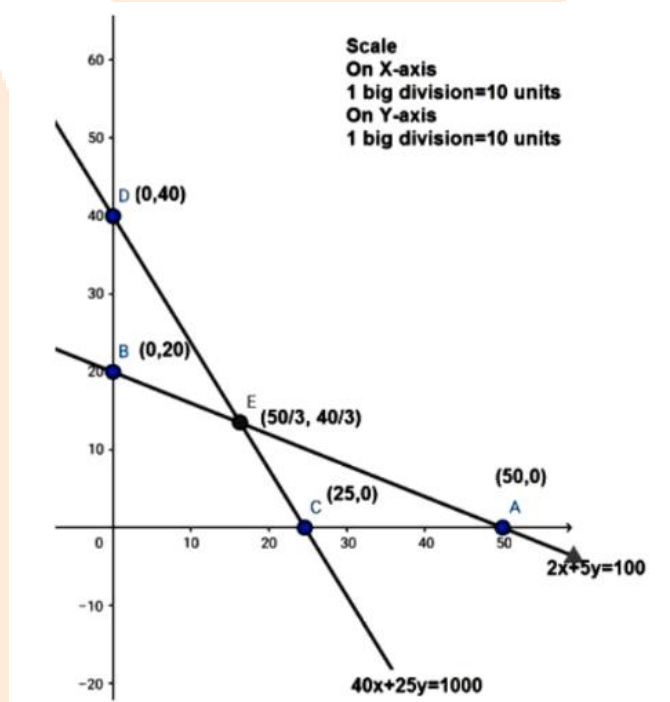
satisfies the $40x + 25y = 1000$. So, the region which contains the origin represents the solution set of the inequation $40x + 25y \leq 1000$

The region represented by $x \geq 0, y \geq 0$:

Since every point in the first quadrant satisfies these inequations. So, the first quadrant is the region represented by the inequations $x \geq 0$ and $y \geq 0$.

The feasible region determined by the system of constraints

$2x + 5y \leq 100, 40x + 25y \leq 1000, x \geq 0$ and $y \geq 0$ are as follows



The corner points are O (0, 0), B (0, 20), E (50/3, 40/3), (50, 0), and C (25, 0). The value of Z at these corner points are as follows:

Corner Points	$Z = x + y$
(0,0)	0
(0,20)	20

$(50/3, 40/3)$	30
$(25, 0)$	25

The maximum value of Z is 30 which is attained at E .

Thus, the maximum distance travelled by the young man is 30kms, if he drives $\frac{50}{3}$ km at a speed of 25km/hr and $\frac{40}{3}$ km at a speed of 40km/hr.

9. A producer has 20 and 10 units of labour and capital respectively which he can use to produce two kinds of goods X and Y. To produce one unit of X, 2 units of capital and 1 unit of labour is required. To produce one unit of Y, 3 units of labour and one unit of capital is required. If X and Y are priced at Rs. 80 and Rs. 100 per unit respectively, how should the producer use his resources to maximize the total revenue?

Ans:

Let x_1 and y_1 units of goods x and y were produced respectively. Number of units of goods cannot be negative.

Therefore,

$$x_1, y_1 \geq 0$$

To produce one unit of x , 2 units of labour and for one unit of y , 1 units of labour are required $2x_1 + y_1 \leq 20$ To produce one unit of x , 3 units of capital is required and 1 unit of capital is required to produce one unit of y $3x_1 + y_1 \leq 10$

If x and y are priced at Rs 80 and Rs 100 per unit respectively,

Therefore, cost of x_1 and y_1 units of goods x and y is Rs $80x_1$ and Rs $100y_1$ respectively.

Total revenue = $Z = 80x_1 + 100y_1$ which is to be maximized.

Thus, the mathematical formulation of the given linear programming problem is

$$\text{Max } Z = 80x_1 + 100y_1$$

subject to $2x_1 + y_1 \leq 20$

$$3x_1 + y_1 \leq 10$$

$$x, y \geq 0$$

First, we will convert inequations into equations as follows:

$$2x_1 + y_1 = 20, 3x_1 + y_1 = 10, x = 0 \text{ and } y = 0$$

Region represented by $2x_1 + y_1 \leq 20$:

The line $2x_1 + y_1 = 20$ meets the coordinate axes at $A(10,0)$ and $B(0,20)$ respectively.

By joining these points, we obtain the line

$$2x_1 + y_1 = 20.$$

Clearly $(0,0)$ satisfies the $2x_1 + y_1 = 20$. So, the region which contains the origin represents the solution set of the inequation $2x_1 + y_1 \leq 20$

Region represented by $3x_1 + y_1 \leq 10$:

The line $3x_1 + y_1 = 10$ meets the coordinate axes at

$C\left(\frac{10}{3}, 0\right)$ and $D(0,10)$ respectively.

By joining these points, we obtain the line $3x_1 + y_1 = 10$.

Clearly $(0,0)$ satisfies the inequation $3x_1 + y_1 \leq 10$.

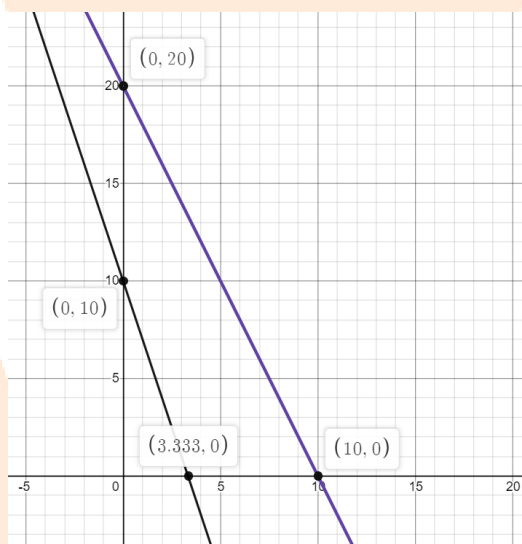
So, the region which contains the origin represents the solution

set of the inequation $3x_1 + y_1 \leq 10$

Region represented by $x_1 \geq 0$ and $y_1 \geq 0$:

Since, every point in the first quadrant satisfies these inequations. So, the first quadrant is the region represented by the inequations $x \geq 0$, and $y \geq 0$.

The feasible region determined by the system of constraints $2x_1 + y_1 \leq 20, 3x_1 + y_1 \leq 10, x \geq 0$ and $y \geq 0$ are as follows.



The maximum value of Z is 760 which is attained at $E(2, 6)$.

Thus, the maximum revenue is Rs 760 obtained when 2 units of x and 6 units of y .

10. A factory owner purchases two types of machines A and B for his factory. The requirements and limitations for the machines are as follows:

Machine	Area Occupied	Labor Force	Daily Outputs
A	1000 m ²	12 Men	60
B	1200 m ²	8 Men	40

He has maximum area of 9000 m² available and 72 skilled laborer's who can operate both the machines. How many machines of each type should he buy to maximize the daily output?

Ans: Let the owner buys x machines of type A and y machines of type B.

Then

$$1000x + 1200y \leq 9000 \quad \dots(i)$$

$$12x + 8y \leq 72 \quad \dots(ii)$$

Objective function is to be maximize $z = 60x + 40y$ From (i)

$$10x + 12y \leq 90$$

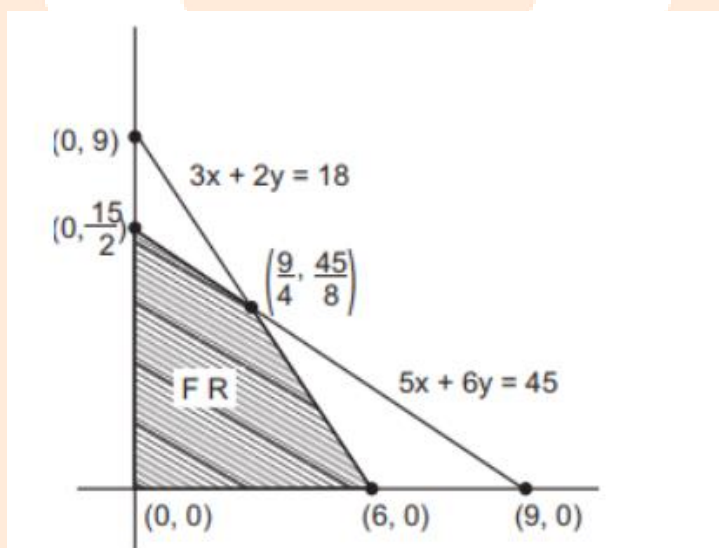
or

$$5x + 6y \leq 45$$

$$3x + 2y \leq 18$$

...(iii) [from (ii)]

We plot the graph of inequations shaded region in the feasible solutions (iii) and (iv).



The shaded region in the figure represents the feasible region which is bounded.

Let us now evaluate Z at each corner point.

at $(0,0)$ Z is $60 \times 0 + 40 \times 0 = 0$

Z at $(0, \frac{15}{2})$ is $60 \times 0 + 40 \times \frac{15}{2} = 300$

Z at $(6,0)$ is $60 \times 6 + 40 \times 0 = 360$

Z at $(\frac{9}{4}, \frac{45}{8})$ is $60 \times \frac{9}{4} + 40 \times \frac{45}{8} = 135 + 225 = 360$

$\Rightarrow \max. Z = 360$

Therefore, there must be either $x=6, Y=0$ or $x=9/4, y=45/8$ but second case is not possible as x and y are whole numbers.

Hence there must be 6 machines of type A and no machine of type B is required for maximum daily output.

11. A manufacturer makes two types of cups A and B. Three machines are required to manufacture the cups and the time in minutes required by each in as given below:

Types of Cups	Machine		
	1	2	3
A	12	18	6
B	6	0	9

Each machine is available for a maximum period of 6 hours per day. If the profit on each cup A is 75 paise and on B is 50 paise, find how many cups of each type should be manufactured to maximize the profit per day.

Ans:

Types of Cups	Machine		
	1	2	3
A	12	18	6
B	6	0	9

Let number of tea-cups A produced be x and number of tea-cups B produced be y

$\therefore$ It is to maximize $Z = 1.50x + 1.y$

i.e., $\text{Max } Z = 1.50x + y$

Subject to $12x + 6y \leq 360$

$$18x + 0.y \leq 360$$

And $6x + 9y \leq 360$

where $x, y \geq 0$

$\therefore$ Given problems can be arranged $\text{Max } Z = 1.50x + y$

Subject to $2x + y \leq 60$

$$x \leq 20$$

And $2x + 3y \leq 120$

$$x \geq 0; y \geq 0$$

Consider the equation $2x + y = 60$

when $x = 0$ then $y = 60$

when $y = 0$ then $x = 30$

Table of solutions

X	0	30
Y	60	0

$$2x + 3y = 120$$

when $x = 0$ then $y = 40$

when $y = 0$ then $x = 60$

Table of solutions

X	0	60
Y	40	0

The corner points of the feasible region from the graph are $O(0,0)$, $A(20,0)$, $B(20,20)$, $C(15,30)$ and $D(0,40)$

At $O(0,0)$; $Z = 1.5(0) + 0 = 0$

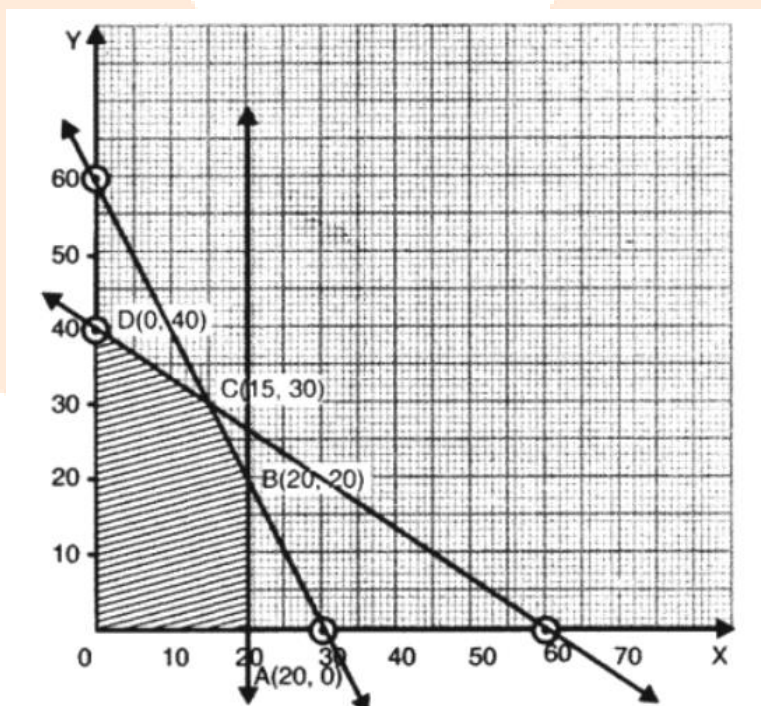
At $A(20,0)$; $Z = 1.5(20) + 0 = 30$

At $B(20,20)$; $Z = 1.5(20) + 20 = 50$

At $C(15,30)$; $Z = 1.5(15) + 30 = 52.50$

At $D(0,40)$; $Z = 1.5(0) + 40 = 40$

Maximum Profit (Z) = Rs.52.50 at $C(15,30)$



Cup A: 15; Cup B: 30

12. A company produces two types of belts A and B. Profits on these belts are Rs. 2 and Rs. 1.50 per belt respectively. A belt of type A requires twice as much time as belt of type B. The company can produce almost 1000 belts of type B per day. Material for 800 belts per day is available. Almost 400 buckles for belts of type A and 700 for type B are available per day. How much belts of each type should the company produce so as to maximize the profit?

Ans:

Let the company produces x belts of types A and y belts of type B. Number of belts cannot be negative.

Therefore, $x, y \geq 0$.

It is given that leather is sufficient only for 800 belts per day (both A and B combined).

Therefore,

$$x + y \leq 800$$

It is given that the rate of production of belts of type B is 1000 per day. Hence the time taken to produce y belts of type B is $\frac{y}{1000}$

And, since each belt of type A requires twice as much time as a belt of type B, the rate of production of belts of type A is 500 per day and therefore, total time taken to produce x belts of type A is $\frac{x}{500}$

Thus, we have,

$$\frac{x}{500} + \frac{y}{1000} \leq 1$$

Or, $2x + y \leq 1000$

Belt A requires fancy buckle and only 400 fancy buckles are available for this per day.

$$x \leq 400$$

For Belt of type B only 700 buckles are available per day.

$$y \leq 700$$

profits on each type of belt are Rs 2 and Rs 1.50 per belt, respectively.

Therefore, profit gained on x belts of type A and y belts of type B is Rs $2x$ and Rs $1.50y$ respectively.

Hence, the total profit would be $\text{Rs}(2x + 1.50y)$. Let Z denote the total profit

$$z = 2x + 1.50y$$

Thus, the mathematical formulation of the given linear programming problem is;

$$\text{Max } Z = 2x + 1.50y \text{ subject to}$$

$$x + y \leq 800$$

$$2x + y \leq 1000$$

$$x \leq 400$$

$$y \leq 700$$

First, we will convert these inequations into equations as follows:

$$x + y = 800$$

$$2x + y = 1000$$

$$x = 400$$

$$y = 700$$

Region represented by $x + y = 800$

The line $x + y = 800$ meets the coordinate axes at $A(800,0)$ and $B(0,800)$ respectively. By joining these points, we obtain the line $x + y = 800$. Clearly $(0,0)$ satisfies the $x + y \geq 800$.

So, the region which contains the origin represents the solution set of the inequation $x + y \geq 800$.

Region represented by $2x + y \geq 1000$

The line $2x + y = 1000$ meets the coordinate axes at $C(500,0)$ and $D(0,1000)$ respectively.

By joining these points, we obtain the line $2x + y = 1000$. Clearly $(0,0)$ satisfies the $2x + y \geq 1000$. So, the region which contains the origin represents the solution set of the inequation $2x + y \geq 1000$.

Region represented by $x \leq 400$

The line $x = 400$ will pass through $(400,0)$. The region to the left of the line $x = 400$ will satisfy the inequation $x \leq 400$.

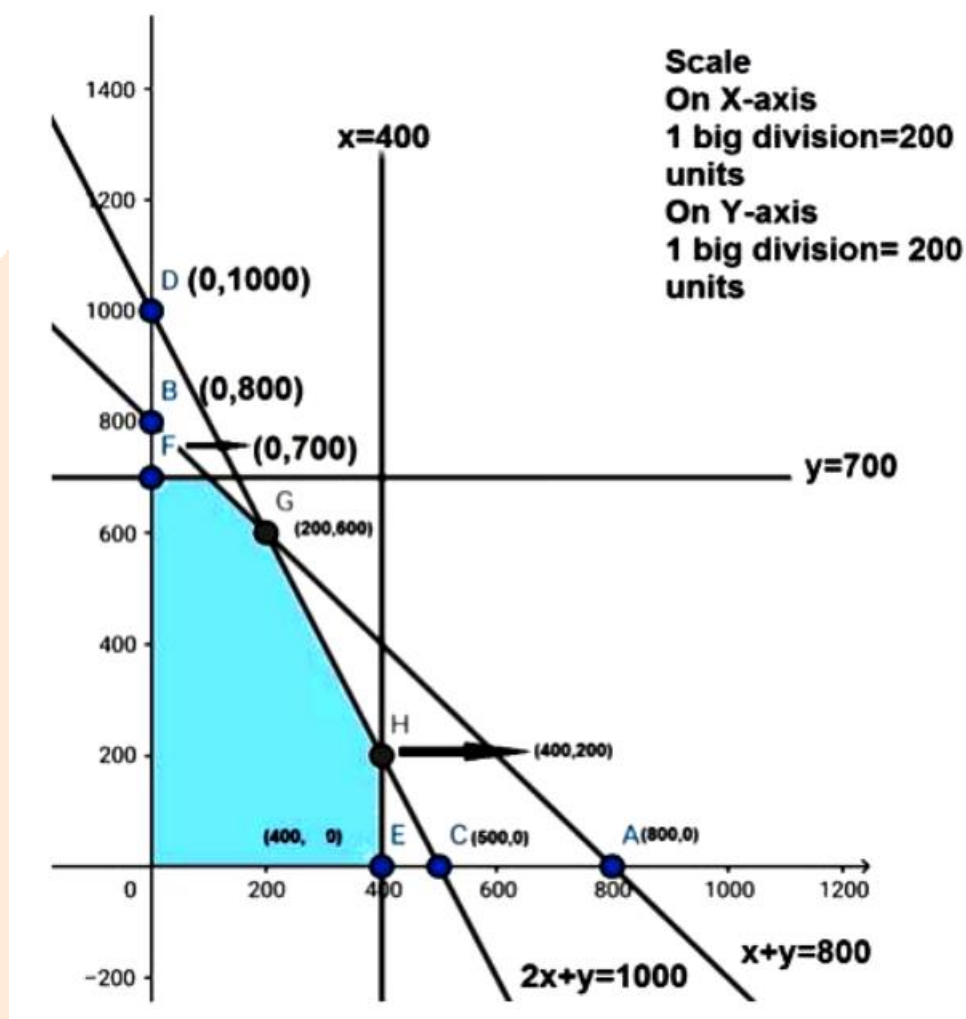
Region represented by $y \leq 700$

The line $y = 700$ will pass through $(0,700)$. The region to the left of the line $y = 700$ will satisfy the inequation $y \leq 700$.

Region represented by $x \geq 0, y \geq 0$:

Since, every point in the first quadrant satisfies these inequations. So, the first quadrant is the region represented by the inequations $x \geq 0$ and $y \geq 0$.

The feasible region determined by the system of constraints $x + y \leq 800$, $2x + y \leq 1000$, $x \leq 400$, $y \leq 700$



The corner points are $F(0, 700)$, $G(200, 600)$, $H(400, 200)$, $E(400, 0)$. The values of Z at these corner points are as follows.

Corners Points	$Z = 2x + 1.5y$
$F(0, 700)$	1050
$G(200, 600)$	1300
$H(400, 200)$	1100

E (400,0)	800
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The maximum value of Z is 1300 which is attained at $G (200,600)$. Thus, the maximum profit obtained is Rs 1300 when 200 belts of type A and 600 belts of type B are produced.

13. Two Godowns X and Y have a grain storage capacity of 100 quintals and 50 quintals respectively. Their supply goes to three ration shop A, B and C whose requirements are 60, 50 and 40 quintals respectively. The cost of transportation per quintals from the godowns to the shops are given in following table:

From	Cost of transformation	
	X	Y
A	6	4
B	3	2
C	2.50	3

How should the supplies be transported to ration shops from godowns to minimize the transportation cost?

Ans:

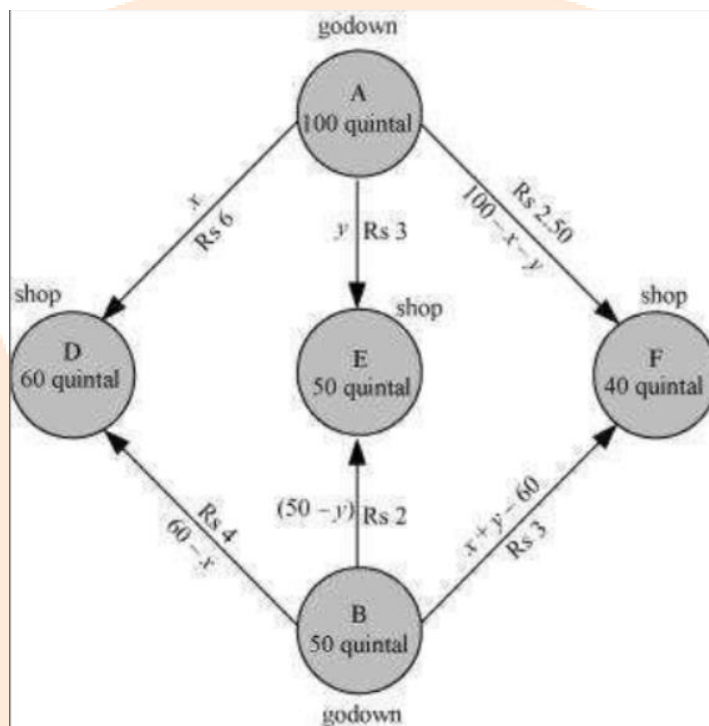
Let go down A supply x and y quintals of grain to the shops D and E respectively.

Then, $(100 - x - y)$ will be supplied to shop F.

The requirement at shop D is 60 quintals since x quintals are transported from godown A. Therefore, the remaining $(60 - x)$ quintals will be transported from godown B.

Similarly, $(50 - y)$ quintals and $40 - (100 - x - y) = (x + y - 60)$ quintals will be transported from godown B to shop E and F respectively.

The given problem can be represented diagrammatically as follows.



$$x \geq 0, y \geq 0, \text{ and } 100 - x - y \geq 0$$

$$\Rightarrow x \geq 0, y \geq 0, \text{ and } x + y \leq 100$$

$$60 - x \geq 0, 50 - y \geq 0, \text{ and } x + y - 60 \geq 0$$

$$\Rightarrow x \leq 60, y \leq 50, \text{ and } x + y \geq 60$$

Total transportation cost z is given by,

$$z = 6x + 3y + 2.5(100 - x - y) + 4(60 - x) + 2(50 - y) + 3(x + y - 60)$$

$$= 6x + 3y + 250 - 2.5x - 2.5y + 240 - 4x + 100 - 2y + 3x + 3y - 180$$

$$= 2.5x + 1.5y + 410$$

The given problem can be formulated as Minimize $z = 2.5x + 1.5y + 410 \dots (1)$

subject to the constraints,

$$x + y \leq 100 \dots\dots\dots(\text{II})$$

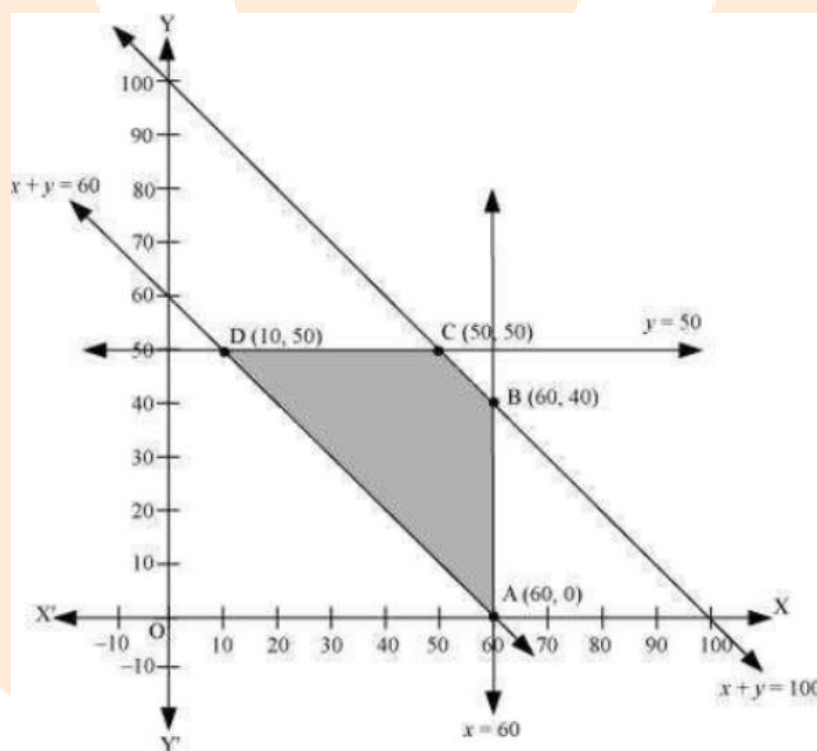
$$x \leq 60 \dots\dots\dots(\text{III})$$

$$y \leq 50 \dots\dots\dots(\text{iv})$$

$$x + y \geq 60 \dots\dots\dots(\text{v})$$

$$x, y \geq 0 \dots\dots\dots(\text{vi})$$

The feasible region determined by the system of constraints is as follows.



The corner points are A (60, 0), B (60, 40), C (50, 50), and D (10, 50). The values of z at these corner points are as follows.

Corner points	$Z = 2.5x + 1.5y + 410$
---------------	-------------------------

A (60,0)	560
B (60,40)	620
C (50,50)	610
D (10,50)	510

The minimum value of z is 510 at (10, 50).

Thus, the amount of grain transported from A to D, E, and F is 10 quintals, 50 quintals, and 40 quintals respectively and from B to D, E, and F is 50 quintals, 0 quintals, and 0 quintals respectively.

14. An Aero plane can carry a maximum of 200 passengers. A profit of Rs. 400 is made on each first-class ticket and a profit of Rs. 300 is made on each second-class ticket. The airline reserves at least 20 seats for first class. However at least four times as many passengers prefer to travel by second class than by first class. Determine, how many tickets of each type must be sold to maximize profit for the airline.

Ans:

Let there be x tickets of executive class and y tickets of economy class. Let Z be net profit of the airline.

Here, we have to maximize z .

Now

$$Z = 500x \times 80/100 + y \times 75/100$$

$$Z = 400x + 300y \dots(I)$$

According to question

$$x \geq 20 \dots(ii)$$

Also

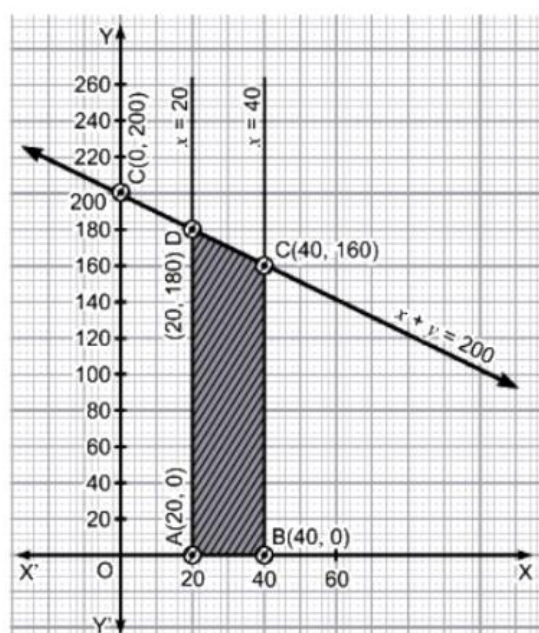
$$x + y \leq 200 \dots (iii)$$

$$\Rightarrow x + 4x \leq 200$$

$$\Rightarrow 5x \leq 200$$

$$\Rightarrow x \leq 40 \dots (iv)$$

Shaded region is feasible region having corner points
A (20, 0), B (40,0) C (40, 160), D (20,180)



Now value of Z is calculated at corner point as

Corners Points	$Z = 400x + 300y$
(20,0)	8,000
(40,0)	16,000
(40,160)	64,000
(20,180)	60,000

Hence, 40 tickets of executive class and 160 tickets of economy class should be sold to maximize the net profit of the airlines.

Yes, more passengers would prefer to travel by such an airline, because some amount of profit is invested for welfare fund.

15. A diet for a sick person must contain at least 4000 units of vitamins, 50 units of minerals and 1400 units of calories. Two foods A and B are available at a cost of Rs. 5 and Rs. 4 per unit respectively. One unit of food A contains 200 unit of vitamins, 1 unit of minerals and 40 units of calories whereas one unit of food B contains 100 units of vitamins, 2 units of minerals and 40 units of calories. Find what combination of the food A and B should be used to have least cost but it must satisfy the requirements of the sick person.

Ans:

We have,

Let x units of food A

y units of food B

Where $x \geq 0, y \geq 0$

We are to minimize

$$Z = 5x + 4y$$

Subject to constraints

$$200x + 100y \geq 4000$$

$$2x + y \geq 40 \quad \dots(1)$$

$$x + 2y \geq 50 \quad \dots(2)$$

$$40x + 40y \geq 1400$$

$$x + y \geq 35 \quad \dots\dots(3)$$

Where,

$$x \geq 0, y \geq 0$$

From equation (1) to and we get,

$$2x + y = 40$$

If $x=0$ then $y = 40$ OY in $L (0, 40)$

If $y=0$ then $x = 20$ OX in $A (20, 0)$

From equation (2) to and we get,

$$X + 2y = 50$$

If $x=0$ then $y = 25$ OY in $M (0, 25)$

If $y=0$ then $x = 50$ OX in $B (50, 0)$

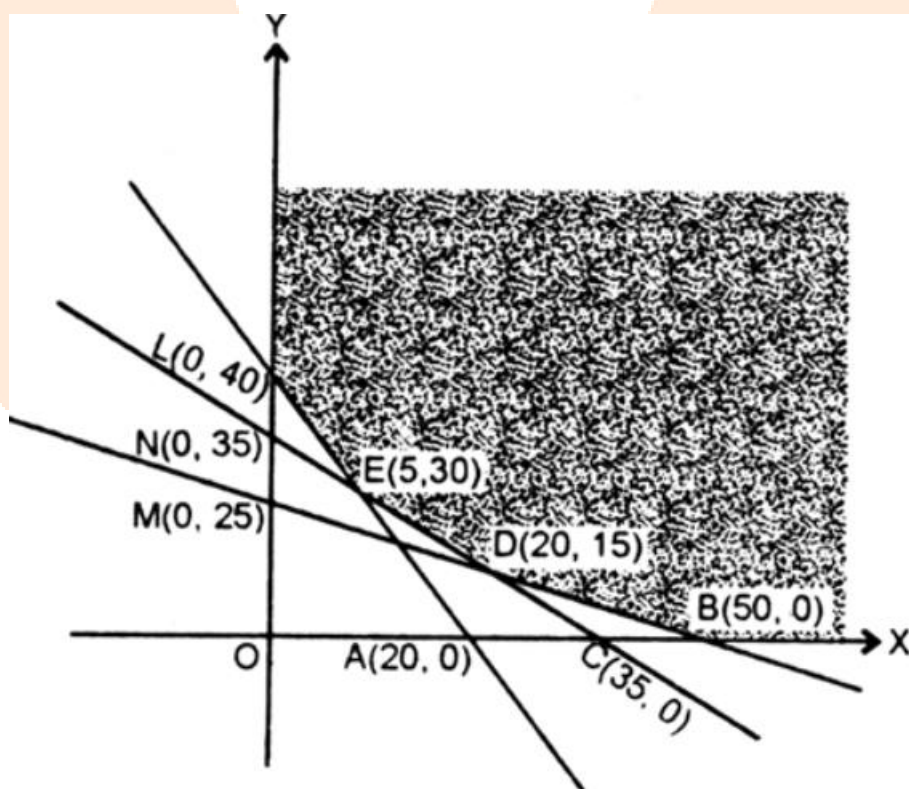
From equation (3) to and we get,

$$X + y = 35$$

If $x=0$ then $y = 35$ OY in $N (0, 35)$

If $y=0$ then $x = 35$ OX in $C (35, 0)$

Shaded region is the feasible region, which is unbounded.



Then,

Corner point is

$B (50, 0)$

$D (20, 15)$

$E (5, 30)$

$L (0, 40)$

At point $B (50, 0) \Rightarrow Z = 5(50) + 4(0) = 250$

At point $D (20, 15), Z = 5(20) + 4(15) = 100 + 60 = 160$

At point $E (5, 30), Z = 5(5) + 4(30) = 25 + 120 = 145$

At point $L (0, 40), Z = 5(0) + 4(40) = 0 + 160 = 160$

Then at least cost = Rs145 at $(5, 30)$

Food A: 5 units, Food B: 30 units.